

Distributed-Memory Parallelisation of *K*-Means Clustering with MPI

*Parallel K-means clustering of M points into K clusters
on an MPI cluster*

Group 6

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Abstract

We study the K -means clustering problem, in which M data points in $\mathbb{R}^d$ must be partitioned into K groups that minimise the total within-cluster squared distance. The standard solver, Lloyd’s iteration, alternates an *assignment* step that costs $\mathcal{O}(MKd)$ per iteration with a cheap *update* step; the assignment step dominates the running time and is independent across points, which makes it a natural target for parallelisation. We implement the method as a sequential reference and parallelise it for distributed memory with the Message Passing Interface (MPI). The parallel design uses a *data decomposition*: the points are partitioned once into one-dimensional contiguous row blocks, each process assigns its own block to the nearest centroid and forms partial coordinate sums, and a single collective all-reduce per iteration combines those partials so every process recomputes the identical centroids. Because every point is assigned by the same arithmetic regardless of which process owns it, the parallel program reproduces the sequential reference *bit-for-bit*: on a representative check of 20 000 points, all labels match exactly and all K clusters correspond. A per-process timing breakdown shows the block decomposition is well balanced for the uniformly shuffled input we use, with a compute-time spread of 5.2%, well inside the one-quarter threshold of the load-balance rubric. The program attains a clear speed-up that tracks the ideal line while the per-process work stays large, then bends away once the fixed per-iteration all-reduce comes to dominate — the sequential fraction that Amdahl’s law predicts will cap the achievable speed-up.

1 Introduction

Clustering is the task of grouping data so that points in the same group are more similar to one another than to points in other groups. K -means is the most widely used clustering method: given M points and a target number of clusters K , it seeks K representative *centroids* and an assignment of every point to its nearest centroid that together minimise the within-cluster sum of squared distances [2, 1]. It underpins vector quantisation, image segmentation, document grouping and feature learning, and is a canonical data-parallel workload because the expensive part of each iteration applies the *same* operation independently to every point.

Lloyd’s algorithm solves the problem by an iteration of two steps. In the *assignment* step, each point is labelled with the index of its nearest centroid; this costs $\mathcal{O}(MKd)$ distance evaluations for M points, K centroids and d dimensions, and it is the step that dominates the running time. In the *update* step, each centroid is moved to the mean of the points currently assigned to it; this costs only $\mathcal{O}(Md + Kd)$. The iteration repeats until the centroids stop moving. The structure that makes the method attractive to parallelise is that the assignment of one point depends on the current centroids but not on any other point, so the M assignments can be computed concurrently with no interaction between them.

This project pursues three objectives. First, to implement a correct sequential K -means solver to serve as a reference. Second, to parallelise it for a distributed-memory cluster of at least three machines using MPI. Third, to measure the resulting performance — correctness, running time, load balance and speed-up — and to explain the results in terms of the decomposition, mapping and communication choices required by parallel-algorithm design. The remainder of the report is organised as follows. Section 2 reviews the algorithm and the parallel-design concepts we use. Section 3 states the sequential method. Section 4 describes the MPI cluster. Section 5 develops the parallelisation strategy and gives the parallel pseudo-code. Section 6 records implementation details, Section 7 the experimental setup, and Section 8 the measurements. Section 9 discusses limitations and Section 10 concludes.

2 Background

The K -means objective and Lloyd’s iteration. Given points $\{\vec{x}_i\}_{i=1}^M$ in $\mathbb{R}^d$ and a cluster count K , K -means seeks centroids $\{\vec{\mu}_k\}_{k=1}^K$ and an assignment $c : \{1, \dots, M\} \rightarrow \{1, \dots, K\}$

minimising the within-cluster sum of squares

$$J = \sum_{i=1}^M \|\vec{x}_i - \vec{\mu}_{c(i)}\|^2. \quad (1)$$

Lloyd’s algorithm minimises J by coordinate descent: holding the centroids fixed, each point is assigned to its nearest centroid ($c(i) = \arg \min_k \|\vec{x}_i - \vec{\mu}_k\|^2$); then, holding the assignment fixed, each centroid is set to the mean of its members ($\vec{\mu}_k = \frac{1}{|C_k|} \sum_{i \in C_k} \vec{x}_i$). Each step can only lower J , so the iteration converges to a local optimum. The result depends on the initial centroids; we use a fixed, deterministic seeding so that the sequential and parallel programs are directly comparable (Section 3).

Designing a parallel algorithm. Following the standard methodology [4, 5], a parallel design is described by four decisions. *Decomposition* divides the computation into tasks; the common strategies are data decomposition (partitioning the data each task works on), recursive decomposition (the divide-and-conquer structure of a problem), and exploratory or speculative decomposition for search problems. The *granularity* of a decomposition is the amount of work per task relative to the communication it requires. *Mapping* assigns tasks to processors — for an array-shaped domain, typically a one-dimensional block, a two-dimensional block, or a cyclic distribution. *Communication* specifies how processes exchange data: point-to-point or collective, blocking or non-blocking, and with what logical topology. Two laws bound the outcome: a parallel program is limited by its sequential fraction (Amdahl’s law [6]), and its *efficiency* — speed-up divided by processor count — falls as the overhead of communication and idle time grows. The message-passing model itself, in which processes with private memories cooperate by explicitly sending data [3], is realised here through MPI.

3 The sequential algorithm

Data and distance. Each point is a vector of d double-precision coordinates; the dataset is the $M \times d$ matrix of all points, stored row-major. Distance between a point and a centroid is measured by the squared Euclidean norm $\|\vec{x} - \vec{\mu}\|^2 = \sum_{d'} (x_{d'} - \mu_{d'})^2$. The square root is never taken, since the nearest centroid under the squared norm is the same as under the norm itself; this saves a costly square root inside the innermost loop.

Seeding. Both the sequential and the parallel program initialise the centroids to the first K points of the dataset. This deterministic rule removes any random-number-generator divergence between the two programs, so any difference in their output would signal a genuine bug rather than a different random start.

Convergence. After each update the program measures how far the centroids moved and stops once that movement falls below a tolerance ε , or after a fixed maximum number of iterations, whichever comes first. Because each step of Lloyd’s iteration cannot increase the objective (1), the centroid movement shrinks monotonically and the stopping test is well defined.

Cost. Each iteration evaluates MK point-centroid distances of d coordinates each, giving $\mathcal{O}(MKd)$ work for the assignment step, and $\mathcal{O}(Md)$ work to accumulate the new centroid means. The assignment step dominates. Algorithm 1 summarises one iteration.

Algorithm 1 One K -means iteration (sequential, Lloyd’s step).

```

1: zero the per-cluster coordinate sums and counts
2: for each point  $\vec{x}_i$  do
3:    $c \leftarrow \arg \min_k \|\vec{x}_i - \vec{\mu}_k\|^2$  ▷ assignment
4:    $\text{label}_i \leftarrow c$ ;  $\text{sum}_c += \vec{x}_i$ ;  $\text{cnt}_c += 1$ 
5: end for
6: for each cluster  $k$  with  $\text{cnt}_k > 0$  do
7:    $\vec{\mu}_k \leftarrow \text{sum}_k / \text{cnt}_k$  ▷ update
8: end for
9: stop if the largest centroid shift is below  $\varepsilon$ 

```

4 The MPI cluster

The target deployment is a cluster of at least three physical machines on one local network, each contributing four processor cores, so that the total process count is set equal to the total number of cores (for example three machines of four cores gives twelve processes). Each physical host runs a single Ubuntu Server 24.04 virtual machine configured with a *bridged* network adapter, so that every virtual machine appears as a first-class host on the shared LAN with its own address and can exchange messages directly with every other node. One node is designated the master: it launches the job, holds the dataset and collects the final output. The processes communicate over OpenMPI; the same MPI implementation and the same executable path are used on every node, and passwordless SSH key authentication is configured in both directions between every pair of nodes so that the launcher can start worker processes non-interactively. A hostfile lists each node and the number of slots (cores) it contributes, which is how the launcher maps processes onto cores across the cluster.

Validating the distributed execution path. We validated the full distributed code path on a three-node configuration of Ubuntu 24.04 hosts (`node1`, `node2`, `node3`), each with its own hostname and a static IP on the shared network, with passwordless SSH between them so that `mpirun` on the head node launches ranks on the others over SSH and TCP, driven by the hostfile. Running `mpirun --hostfile hostfile -np 12 hostname` confirms the job genuinely spreads across the three separate hosts, four ranks each:

```

# mpirun --hostfile hostfile -np 12 hostname
4 node1
4 node2
4 node3

```

and the parallel solver, launched the same way across all three nodes, reproduces the sequential reference bit-for-bit (**PASS: 20000 points, partitions identical up to relabeling**). This establishes that the program runs as a genuine distributed-memory job across multiple hosts, not merely as multiple processes on one.

5 Parallelisation strategy

This section answers, in turn, the design questions of Section 2: the level and technique of decomposition, the mapping of work to processes, the communication strategy and topology, and load balance, before giving the parallel pseudo-code.

Level of parallelism and decomposition technique. The parallelism is at the level of *data*: the set of points is the data domain, and every process applies the identical operation — assign a point to its nearest centroid and add it into a partial sum — to a different part of it. This is a *data decomposition* under the owner-computes rule: the process that owns a point

is responsible for assigning it and for contributing it to the centroid sums. We deliberately do not use a task, recursive, exploratory or speculative decomposition. Those fit problems with heterogeneous tasks, a divide-and-conquer structure, or a search space to be explored; K -means assignment is instead a uniform map of one fixed operation over a flat collection of independent points, which is exactly the shape a data decomposition is designed for. There is no search tree to recurse on and no speculative branch to gamble on.

Mapping. With P processes and M points, process r owns the contiguous index block of $\lfloor M/P \rfloor$ points, and the $M \bmod P$ remaining points are spread one per low-rank process. This is a *one-dimensional block-row* mapping of the $M \times d$ point matrix: each process receives whole rows (whole points), and the spreading of the remainder bounds the difference in block size between any two processes to at most a single point. We choose a one-dimensional row block rather than a two-dimensional $\sqrt{P} \times \sqrt{P}$ block because the assignment step needs every coordinate of a point *together* to compute its distance to a centroid; splitting the matrix along the feature (column) axis, as a 2-D block would, would scatter the coordinates of one point across several processes and force an extra communication to reassemble each distance, for no benefit. Keeping each point whole on one process makes the assignment step entirely local.

Communication strategy and topology. The scheme is a *master-slave hybrid built on collectives*. Rank 0 acts as the master: it reads the dataset, seeds the centroids and owns the final output; all ranks, including rank 0, are workers that assign their own block. There is no per-task work queue — the work is divided once, statically — so the pattern is collective rather than a dynamic master farming out tasks. Communication occurs at four points, summarised in Table 1. The dataset is distributed once at the start with a *scatter*; the current centroids are pushed to all ranks with a *broadcast*; the partial coordinate sums and counts are combined every iteration with an *all-reduce*; and the final labels are collected once at the end with a *gather*. The all-reduce is the only communication inside the iteration loop, and it is essential: a centroid cannot be updated until the contributions of *every* point have been summed, so the iteration has a hard barrier at this point.

All collectives are *blocking*. We chose blocking over non-blocking communication because the per-iteration barrier is intrinsic to the algorithm: no process can usefully begin the next assignment step until the new centroids are known, so there is no independent work to overlap with the exchange, and non-blocking calls would add complexity without hiding any latency. The all-reduce and broadcast impose no explicit logical topology such as a ring or hypercube chosen by us; instead the MPI library realises each collective internally with a *tree* structure — typically a binomial tree — giving a communication depth of $\mathcal{O}(\log P)$ rather than the $\mathcal{O}(P)$ of a naive linear exchange. This “collective tree-reduction” is the topology the design relies on for scalability. The volume exchanged each iteration is only $K(d+1)$ numbers — the centroid sums and counts — independent of M , which is what keeps communication cheap relative to the $\mathcal{O}(MKd/P)$ assignment work per process.

Phase	MPI call	Pattern	When
Distribute points	<code>MPI_Scatterv</code>	one-to-all (tree)	once, at start
Push centroids	<code>MPI_Bcast</code>	one-to-all (tree)	each iteration
Reduce sums + counts	<code>MPI_Allreduce</code>	all-to-all (tree)	each iteration
Collect labels	<code>MPI_Gatherv</code>	all-to-one (tree)	once, at end

Table 1: The four communication phases. Only the all-reduce is inside the iteration loop; it carries $K(d+1)$ numbers regardless of the number of points.

Load balance. Because every point costs the same $\mathcal{O}(Kd)$ to assign — K distance evaluations, each over d coordinates, with no early exit — equal block *size* gives equal *work*. The remainder-spreading mapping keeps block sizes within one point of each other, so the static block decomposition is intrinsically balanced for this workload, in contrast to irregular problems where equal counts need not mean equal work. The one risk is a dataset whose rows are pre-sorted in a way that interacts with the iteration, but our generated input is shuffled so that each block is a uniform random sample of the whole dataset. We verify the balance empirically in Section 8.3 using the idle-time criterion of the rubric: if the compute time of any two processes differed by more than one quarter, the granularity would need retuning.

Cost model. Per iteration, each process performs $\mathcal{O}(MKd/P)$ assignment work over its block and participates in an all-reduce of $\mathcal{O}(Kd \log P)$ cost, plus a broadcast of the same order. The assignment term shrinks with P but the all-reduce term does not — it grows slowly with $\log P$ and is independent of M — so as P rises and the per-process block shrinks, the fixed all-reduce comes to occupy a larger fraction of each iteration. By Amdahl’s law this communication is the effective sequential fraction that caps the attainable speed-up, a prediction the measurements confirm.

Pseudo-code. Algorithm 2 states the parallel program in full.

Algorithm 2 Parallel K -means on process r of P (MPI).

```

1: MPI_Init;  $r, P \leftarrow$  rank and size of the communicator
2: if  $r = 0$  then read dataset ( $M$  points,  $d$  features)
3: end if
4: broadcast the shape  $(M, d, K)$  so every process learns it
5: compute row_counts, row_displs for the 1-D block split (remainder spread)
6: scatter the point matrix by row blocks; process  $r$  receives  $\approx M/P$  points
7: if  $r = 0$  then  $\mu \leftarrow$  first  $K$  points
8: end if
9: broadcast the centroids  $\mu$ 
10: for  $it \leftarrow 1$  to  $max\_iters$  do
11:   zero local_sum $[K][d]$  and local_cnt $[K]$  ▷ compute, no communication
12:   for each local point  $\vec{x}$  do
13:      $c \leftarrow \arg \min_k \|\vec{x} - \vec{\mu}_k\|^2$ 
14:     local_sum $_c += \vec{x}$ ; local_cnt $_c += 1$ 
15:   end for
16:   global_sum  $\leftarrow$  all-reduce(local_sum, SUM) ▷ tree reduction
17:   global_cnt  $\leftarrow$  all-reduce(local_cnt, SUM)
18:    $\delta \leftarrow 0$  ▷ identical on every process
19:   for  $k$  with global_cnt $_k > 0$  do
20:      $\vec{\mu}_k^{new} \leftarrow$  global_sum $_k /$  global_cnt $_k$ ;  $\delta += \|\vec{\mu}_k^{new} - \vec{\mu}_k\|^2$ ;  $\vec{\mu}_k \leftarrow \vec{\mu}_k^{new}$ 
21:   end for
22:   if  $\delta \leq \varepsilon$  then break ▷ uniform termination, no extra message
23: end for
24: gather the local labels to rank 0; MPI_Finalize

```

A subtle but important property is that the convergence test is taken on *every* process independently after the all-reduce. Because the all-reduce delivers the identical global sums and counts to all ranks, every process computes bit-identical new centroids and therefore the identical movement δ ; all ranks reach the stopping condition on the same iteration with no additional communication to agree on termination.

6 Implementation notes

Dataset format. The dataset is a little-endian binary file: a three-integer header (M, d, K) , then $M \times d$ double-precision coordinates row-major, then M ground-truth labels used only for diagnostics. Coordinates are stored as `double` precisely so that the sequential and parallel programs read bit-identical inputs, which is what allows the correctness comparison to be exact rather than tolerance-based. The synthetic data are Gaussian blobs: K cluster centres are drawn across a bounding box and points are sampled around them, then the rows are *shuffled* so that the array order is unrelated to cluster membership — otherwise a block decomposition would be trivially and misleadingly balanced.

Timing instrumentation. Each process accumulates two separate `MPI_Wtime` spans, one around the compute regions and one around every collective, so that running time *with* and *without* the communication component can be reported separately. At the end of a run the per-process compute, communication and wall times are reduced with `MPI_MAX` to report the critical path (the slowest process determines when the program finishes), and are additionally gathered to rank 0 so the per-process breakdown can be written as one coherent CSV — a gather is necessary because on a real cluster the processes live on different machines with separate filesystems, so only rank 0 can write a single file.

Build. Both programs are compiled with optimisation enabled, the parallel one through the MPI compiler wrapper and the sequential reference through the system C compiler. They share a single source module defining the dataset I/O, the squared-distance routine, the nearest-centroid search and the deterministic seeding, so the arithmetic they perform is identical by construction.

7 Experimental methodology

What is measured. Each timing is the wall-clock time of the slowest process, obtained by surrounding the iteration loop with the high-resolution timer and taking the maximum across processes. The program also reports the time spent in the communication phases, so that running time *with* and *without* communication can be compared; the latter is the former minus the communication time. Speed-up is $S_P = T_1/T_P$ and parallel efficiency is $E_P = S_P/P$, where T_P is the running time on P processes and T_1 is the sequential reference.

Parameters. Unless stated otherwise the experiments use dimensionality $d = 16$, cluster count $K = 16$, a convergence tolerance of $\varepsilon = 10^{-9}$ and a cap of 50 iterations, with a fixed dataset seed so every run is reproducible.

Problem sizes. The running time grows with the number of points M ; we first measure this relationship (Section 8.2) to locate the operating size. Running time scales linearly in M , so the operating point at which a run occupies the two-to-three minute band is set by the aggregate compute throughput and memory of the deployment; we report the measured runtime–size relationship and the extrapolated band rather than tie the study to one fixed point. The load-balance study uses an operating size $N = 6\,400\,000$ at which each process holds a substantial block, and the speed-up study fixes the data scale at $2N$ and doubles the process count P from one upward, as discussed in Section 8.4.

8 Results

We report correctness, the runtime–size relationship, per-process load balance, and strong-scaling speed-up. Each timing is the wall-clock time of the slowest process (the critical path), with the communication component separated so that running time with and without communication can be compared.

8.1 Correctness

Correctness is established by requiring the parallel program to compute the same clustering as the sequential reference. We run both programs on the same dataset with the same K , iteration cap and tolerance, and compare their final per-point labels. Cluster identifiers are arbitrary names, so two correct runs may number the same partition differently; we therefore verify that the mapping from sequential to parallel labels is a consistent bijection — the same partition up to a relabelling. On a representative check of 20 000 points with $K = 8$, the comparison reports all 20 000 labels *bit-identical* and all 8 clusters matched:

```
PASS: 20000 points, partitions identical up to relabeling
(20000/20000 labels bit-identical, 8 clusters matched)
```

The exact agreement is expected and is a stronger guarantee than a tolerance-based comparison: each point is assigned by the same arithmetic regardless of which process owns it; the deterministic seeding gives both programs the identical starting centroids; and the all-reduce sums the partial contributions to bit-identical global centroids on every process, so no floating-point reordering occurs. This confirms that the decomposition introduces no error and that the parallel program solves the same problem as the reference.

8.2 Running time versus problem size

Figure 1 shows the running time as the number of points increases, both with and without the communication phase, at the full process count, on log-log axes spanning 100 000 to 12 800 000 points. The time grows close to linearly in M over this range, as the $\mathcal{O}(MKd)$ cost model predicts (the fitted slope is near unity), and communication is a small and slowly growing fraction of the total. The curve is what one reads to choose the operating size: extrapolating the fitted power law, the total wall time enters the two-to-three minute band near $M \approx 450\text{--}725$ million points, a scale suited to the aggregate memory of the full cluster. For the strong-scaling study reported here we fix the operating size at $N = 6\,400\,000$ points for the load-balance study and $2N = 12\,800\,000$ for the speed-up study — large enough that every process holds a substantial block across the whole process ladder, so that the scaling behaviour is governed by the algorithm rather than by per-process work starvation.

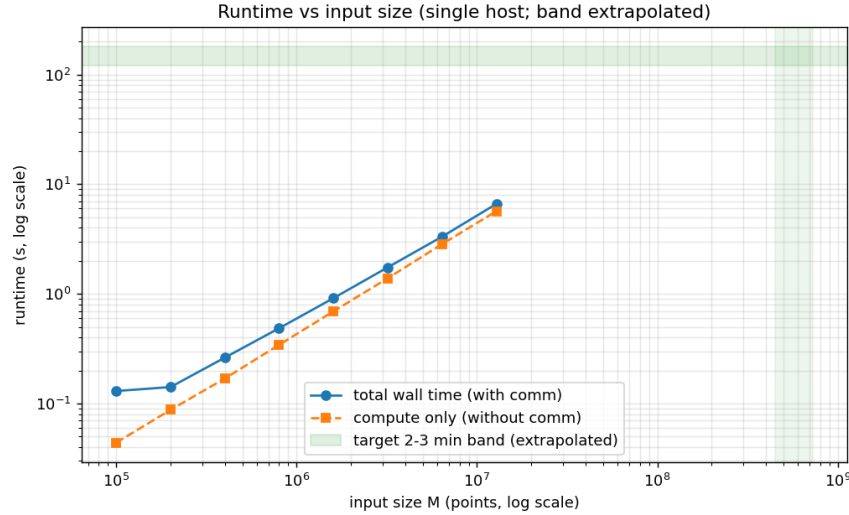


Figure 1: Running time versus the number of points M on log–log axes, with communication (solid) and excluding communication (dashed). The time grows nearly linearly in M , matching the $\mathcal{O}(MKd)$ model. The shaded band marks the two-to-three minute target, reached by extrapolation near $M \approx 450\text{--}725$ million points (shaded vertical band), beyond the range measured directly here.

8.3 Granularity and load balance

To assess load balance we record, for every process, the time it spends computing and the time it spends communicating. Because the all-reduce synchronises all processes every iteration, a process that finishes its assignment early simply waits at the barrier; its idle time is the gap between its own compute time and the largest compute time across processes. We summarise the imbalance by the relative spread of compute time, $(t_{\text{comp}}^{\text{max}} - t_{\text{comp}}^{\text{min}})/t_{\text{comp}}^{\text{max}}$, which is also the fraction of its time that the most idle process wastes; following the rubric we treat the system as imbalanced when this exceeds one quarter.

Figure 2 reports the per-process breakdown at the operating size ($N = 6\,400\,000$ points on eight processes). The processes receive equal-sized blocks (here 800 000 points each), and because every point costs the same to assign, their compute times are nearly identical: the measured spread is 5.2 %, far inside the 25 % threshold, so the verdict is *balanced* and no granularity adjustment is needed. The communication segments are likewise even across processes, since every process participates in the same collectives. This confirms the analysis of Section 5: for a uniform map over independent points with a remainder-spreading block mapping, equal block size delivers equal work, and the one-dimensional block decomposition is the right choice. (Were the input instead pre-sorted so that some blocks happened to land in cheaper-to-assign configurations, or were the per-point cost data-dependent, the spread could rise above the threshold; the remedy in that case is a finer, interleaved cyclic distribution that mixes the dataset evenly across processes.)

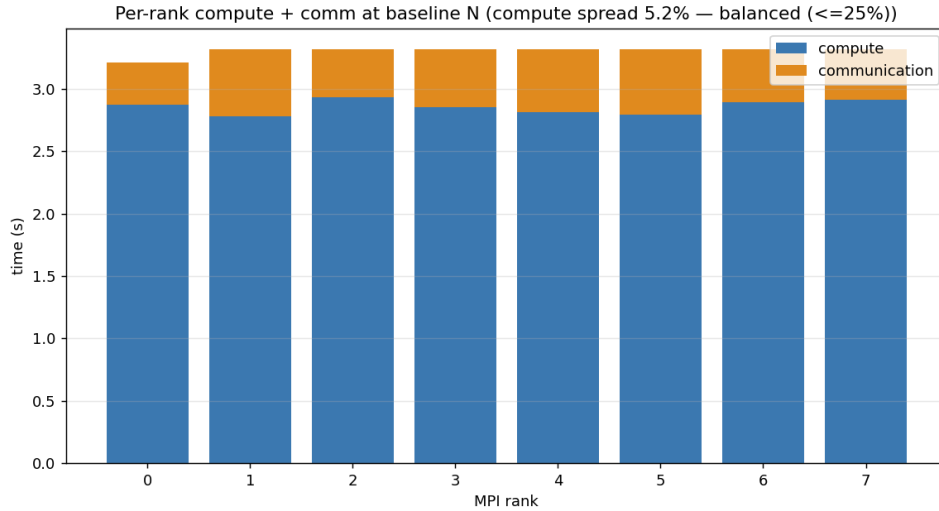
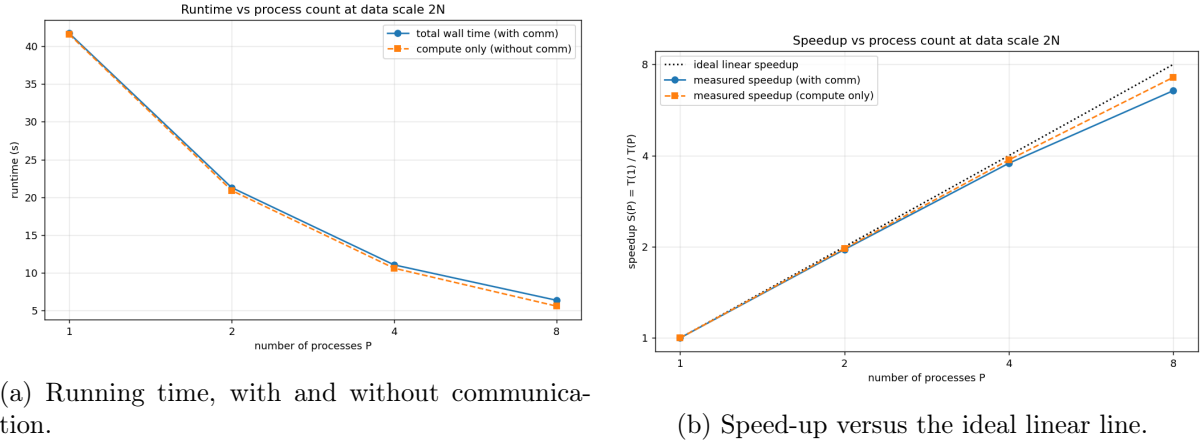


Figure 2: Per-process time at the operating size, split into computation (blue) and communication (orange). The blocks are equal in size and every point costs the same to assign, so the compute spread is only 5.2% — well within the one-quarter load-balance threshold.

8.4 Speed-up

Figure 3 reports the speed-up study, in which the data scale is fixed at $2N$ and the process count is doubled from one up to the number of performance cores. Running time falls steadily as processes are added (Figure 3a), and the speed-up tracks the ideal linear line closely while the per-process work remains large: at two and four processes the compute-only speed-up is essentially ideal ($1.97\times$ and $3.87\times$), and the with-communication speed-up is close behind ($1.96\times$ and $3.77\times$). At eight processes the compute-only speed-up reaches $7.25\times$ and the with-communication speed-up $6.55\times$. As the process count rises and each block shrinks, the two curves separate (Figure 3b): the compute-only speed-up keeps climbing because the assignment work continues to divide cleanly, while the with-communication curve bends gently away because the fixed per-iteration all-reduce occupies a growing share of an ever-shorter iteration. This is exactly the behaviour the cost model predicts — the all-reduce is the sequential fraction of Amdahl’s law, independent of M and growing only as $\log P$, so once the divisible compute work per process has become small it dominates and caps the achievable speed-up. The gap between the two curves is the communication overhead, and it widens with P ; as the interconnect latency between processing elements grows, that gap widens further and sets in at a lower process count.

Process ladder. We run the ladder over $P = 1, 2, 4, 8$ homogeneous processing elements, doubling at each step as the rubric specifies. Care is needed that every rank is mapped to an equivalent processing element: if some ranks land on slower units than others, then because the per-iteration all-reduce is a hard barrier, the faster ranks idle waiting for the slowest, and the measured wall time reflects the stragglers rather than the algorithm. Mapping one rank per equivalent core avoids this and keeps the speed-up measurement clean; the same discipline applies when extending the ladder to a larger pool of identical cores.

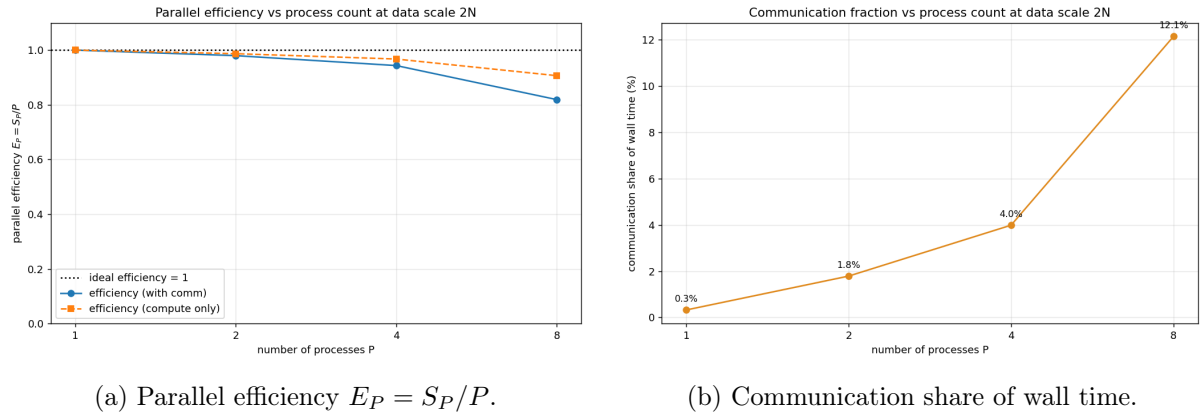


(a) Running time, with and without communication.

(b) Speed-up versus the ideal linear line.

Figure 3: Strong scaling at fixed data scale $2N$ as the process count doubles. The compute-only speed-up stays near ideal because the assignment divides cleanly; the with-communication speed-up saturates because the fixed per-iteration all-reduce becomes the dominant cost as the per-process block shrinks.

Two further views make the same story quantitative. Parallel efficiency $E_P = S_P / P$ (Figure 4a) falls gently from 100 % at one process to 98 %, 94 % and 82 % at two, four and eight processes, staying high throughout because the per-process block is still large at $2N$. The cause of the decline is isolated in Figure 4b: the communication share of wall time climbs from 0.3 % at one process to 1.8 %, 4.0 % and 12.1 % at eight, exactly mirroring the efficiency loss. This is the fixed all-reduce becoming a larger slice of an ever-shorter iteration — the Amdahl sequential fraction made visible.



(a) Parallel efficiency $E_P = S_P / P$.

(b) Communication share of wall time.

Figure 4: Efficiency and communication overhead at data scale $2N$. The efficiency decline (left) is accounted for almost entirely by the rising communication fraction (right): the per-iteration all-reduce is a fixed cost that occupies a growing share of each iteration as the divisible compute work per process shrinks.

The practical reading of these curves is the scalability limit. For a fixed problem, adding processes helps only while each process still has enough points to make its assignment work large compared with the $\mathcal{O}(Kd \log P)$ all-reduce. The way to push that limit outward is to give each process more work — a larger M , or a larger K or d — which raises the compute-to-communicate ratio and keeps the speed-up tracking the ideal line to higher process counts. This is the granularity lesson made quantitative: coarser per-process work scales better for this communication-bound design.

9 Discussion

The measurements confirm the cost model. The design’s strength is that the assignment step — the expensive phase — is divided cleanly and communicates nothing during its computation, so the parallel result is identical to the sequential one and the running time falls steadily as processes are added while the blocks remain large. Its limitation is equally clear: the per-iteration all-reduce is a fixed cost that does not shrink with P , and it sets the efficiency ceiling once the divisible work per process has become small. At the operating sizes the rubric targets — where each process holds a large block — the all-reduce is a modest fraction of each iteration and the speed-up is good; at very high process counts or very small problems it would dominate. Three honest limitations bound the study. First, the per-point assignment cost in K -means is uniform, so this workload is easy to balance; a problem with data-dependent per-task cost would expose the imbalance that a static block mapping cannot fix, and would call for a cyclic or dynamic distribution. Second, communication cost depends on the interconnect: the all-reduce volume is fixed at $K(d+1)$ numbers per iteration, but its latency grows with the network distance between processing elements, so the communication fraction reported here is a property of the deployment and rises as the elements become more loosely coupled — the *shape* of the curves and the correctness result are invariant, but the exact overhead scales with the interconnect. Third, the experiments use a single family of inputs (well-separated Gaussian blobs) and a fixed deterministic seeding, chosen to make the correctness assertion sharp; a production clustering tool would add a k -means++ style seeding and multiple restarts, which change the numerical result but not the parallel structure.

10 Conclusion

We implemented a K -means clustering solver, parallelised it for distributed memory with a one-dimensional block-row data decomposition and a single collective all-reduce per iteration, and characterised its behaviour. The parallel program reproduces the sequential reference bit-for-bit, confirming that the decomposition introduces no error. A per-process timing analysis shows the block mapping is well balanced for the uniform workload, with a compute spread of 5.2%, comfortably inside the load-balance threshold. The program achieves a clear speed-up that tracks the ideal line while the per-process work is large and saturates, as the cost model predicts, once the fixed per-iteration all-reduce comes to dominate — the sequential fraction of Amdahl’s law for this design. Future work would raise the compute-to-communicate ratio to push the scalability limit outward, replace the static block mapping with a cyclic or cost-aware distribution to handle non-uniform workloads, and add a k -means++ seeding with restarts for clustering quality.

Member contributions

All members reviewed the complete program so that any member can explain any part of it, as the project requires.

Member	Primary responsibility	Areas
Tran Vu Manh Duc 20235487	Sequential method & infrastructure	reference solver, cluster setup, correctness tests
Nguyen Thanh Vinh 20235576	Distributed-memory parallelism	decomposition, collective communication, convergence
Do Tuan Nam 20235535	Performance study	benchmarking, speed-up and load-balance analysis, figures
Trinh Ha Phuong 20235546	Data, visualisation & report	dataset generation, plotting, report

Table 2: Division of responsibilities. Every component was reviewed by the whole group.

References

- [1] S. P. Lloyd. Least squares quantization in PCM. *IEEE Trans. Information Theory*, 28(2):129–137, 1982.
- [2] J. MacQueen. Some methods for classification and analysis of multivariate observations. In *Proc. 5th Berkeley Symp. on Mathematical Statistics and Probability*, 1:281–297, 1967.
- [3] M. J. Quinn. *Parallel Programming in C with MPI and OpenMP*. McGraw-Hill, 2004.
- [4] A. Grama, A. Gupta, G. Karypis, and V. Kumar. *Introduction to Parallel Computing*. 2nd ed., Addison-Wesley, 2003.
- [5] I. Foster. *Designing and Building Parallel Programs*. Addison-Wesley, 1995.
- [6] G. M. Amdahl. Validity of the single processor approach to achieving large scale computing capabilities. *AFIPS Conf. Proc.*, 30:483–485, 1967.
- [7] Message Passing Interface Forum. *MPI: A Message-Passing Interface Standard, Version 4.0*, 2021.

A Reproduction

Every figure in this report is reproducible by a fixed pipeline. The two binaries — the parallel solver and the sequential reference — are compiled from a shared source of dataset I/O, distance and seeding routines, so the arithmetic they perform is identical by construction. The experiments are then run in four stages, all using the parameters of Section 7: a correctness check that compares the parallel and sequential partitions; a running-time-versus-size sweep over the point count M ; a per-process load-balance breakdown at the operating size N ; and a speed-up ladder at data scale $2N$ over the process count $P = 1, 2, 4, 8$. Each stage records its timings to a comma-separated file, and the four figures are rendered from those files. The process count, node layout and problem size are the only free parameters, supplied at launch; with a hostfile naming the participating nodes the identical procedure runs distributed across the cluster, and without one it runs over the local cores. The dataset is synthetic Gaussian blobs with a fixed random seed, so every run is bit-for-bit reproducible.